

Q4. MAR-Valid Modeling Strategies for Continuous and Binary Outcomes

The analyses in this assignment rely on the assumption that missingness is Missing At Random (MAR). Although MAR cannot be formally tested, the exploratory missingness analysis demonstrated monotone dropout and clear associations between missingness and observed data, making MAR a plausible working assumption. Under MAR, different modeling frameworks require different strategies to ensure valid inference. This section presents and justifies MAR-valid versions of the models used for the continuous and binary outcomes.

Q4.1 MAR-Valid Linear Mixed Model for the Continuous Outcome

For the continuous BPRS outcome, inference was based on a linear mixed-effects model (LMM) estimated using restricted maximum likelihood. The model includes a patient-specific random intercept and a random slope for time, allowing for heterogeneity in both baseline levels and longitudinal trajectories.

Because the LMM is likelihood-based, valid inference under MAR follows from the ignorability of the missingness mechanism: under MAR, the likelihood for the observed data can be maximized without explicitly modeling the dropout process. Consequently, no additional correction for missingness—such as weighting or imputation—is required, provided that the mean structure and random-effects structure are correctly specified.

The strong improvement in model fit when including a random slope for time further supports the adequacy of the selected model. The final LMM therefore provides a MAR-valid analysis of the continuous outcome and serves as the primary inferential framework for BPRS.

Q4.2 MAR-Valid Random-Effects Model for the Binary Outcome (GLMM)

For the binary outcome derived from CDRSB, a generalized linear mixed model (GLMM) with a logit link was fitted, including a patient-specific random intercept and a random slope for time. As with the LMM, the GLMM is estimated via maximum likelihood and therefore benefits from the same ignorability property under MAR.

Under the MAR assumption, likelihood-based inference for the GLMM remains valid without explicitly modeling the missingness mechanism. The fitted GLMM captures substantial subject-specific heterogeneity and yields conditional (subject-specific) effect estimates. Importantly, the substantive conclusions from the GLMM are consistent with those obtained from the marginal analysis, reinforcing the robustness of the findings.

Thus, the GLMM constitutes a MAR-valid analysis of the binary outcome without the need for additional missing-data corrections.

Q4.3 MAR-Valid Marginal Models for the Binary Outcome

In contrast to likelihood-based mixed models, standard GEE does not automatically yield valid inference under MAR, as it relies on estimating equations rather than a full likelihood. Therefore, additional steps are required to obtain MAR-valid marginal inference for the binary outcome.

Two MAR-based approaches were considered: inverse probability weighted GEE (IPW-GEE) and multiple imputation followed by GEE (MI-GEE).

Inverse Probability Weighted GEE (IPW-GEE).

Inverse probability weights were derived from a logistic regression model for the observation process, including time and baseline covariates. Stabilized weights were used, and extreme values were truncated at the 99th percentile to ensure numerical stability.

Table. 3. Summary statistics of stabilized inverse probability weights (w) and trimmed stabilized weights (w(trim)) used in the IPW-GEE analysis;

Variable	N	Mean	Std Dev	Minimum	50th Pctl	95th Pctl	99th Pctl	Maximum
w	6220	0.9208203	0.8574226	0.3740367	0.8753635	1.3658753	3.5061286	35.2571396
w_trim	6220	0.8844342	0.4163296	0.3740367	0.8753635	1.3658753	3.5061286	3.5061286

The weighted GEE was then fitted using the same marginal mean structure as the unweighted analysis. The resulting IPW-GEE estimates are valid under MAR and yielded conclusions that were qualitatively identical to those from the unweighted GEE.

Table 8. Parameter estimates from the inverse probability weighted generalized estimating equation (IPW-GEE) model for the binary outcome, with empirical (robust) standard errors and 95% confidence intervals.

Parameter		Estimate	Standard Error	95% Confidence Limits		Z	Pr > Z
Intercept		-0.5363	1.0847	-2.6624	1.5897	-0.49	0.6210
time		1.2139	0.0570	1.1022	1.3257	21.29	<.0001
time2		-0.1580	0.0104	-0.1784	-0.1377	-15.24	<.0001
age		-0.0007	0.0064	-0.0132	0.0118	-0.10	0.9165
sex	0	0.0713	0.0576	-0.0417	0.1842	1.24	0.2162
sex	1	0.0000	0.0000	0.0000	0.0000	.	.
bmi		-0.0096	0.0125	-0.0341	0.0148	-0.77	0.4398
adl		0.0127	0.0110	-0.0088	0.0343	1.16	0.2471
abpet0		0.1530	0.1045	-0.0518	0.3577	1.46	0.1431
taupet0		0.0120	0.5216	-1.0103	1.0342	0.02	0.9817

Multiple Imputation followed by GEE (MI-GEE).

As an alternative MAR-based strategy, missing binary outcomes were multiply imputed using a fully conditional specification with a logistic imputation model. The marginal GEE was fitted to each imputed dataset, and estimates were pooled using Rubin's rules (Table 9).

Table 9. Rubin-pooled parameter estimates from the multiple-imputation generalized estimating equation (MI-GEE) analysis for the binary outcome, with standard errors and confidence intervals accounting for between- and within-imputation variability.

Obs	Parameter	Nimpute	Param	Estimate	StdErr	LCLMean	UCLMean	DF	Min	Max	Theta0	tvalue	Probt
1	abpet0	10	Estimate	0.108497	0.121354	-0.1374	0.3544	37.141	-0.028470	0.255970	0	0.89	0.3771
2	adl	10	Estimate	0.004542	0.011460	-0.0180	0.0271	214.94	-0.001397	0.016267	0	0.40	0.6922
3	age	10	Estimate	-0.001613	0.006323	-0.0141	0.0108	298.75	-0.005031	0.003677	0	-0.26	0.7988
4	bmi	10	Estimate	-0.011387	0.013082	-0.0373	0.0146	100.21	-0.021504	0.000615	0	-0.87	0.3851
5	intercept	10	Estimate	-0.304479	1.233958	-2.9206	2.3117	15.981	-1.578354	1.071349	0	-0.25	0.8082
6	sex=0	10	Estimate	0.053210	0.067508	-0.0825	0.1890	47.875	-0.023084	0.106670	0	0.79	0.4345
7	sex=1	10	Estimate	0	0	-	-	-	0	0	0	-	-
8	taupet0	10	Estimate	0.042026	0.601974	-1.2614	1.3455	12.717	-0.618798	0.881702	0	0.07	0.9454
9	time	10	Estimate	1.202912	0.051352	1.1019	1.3039	317.7	1.177658	1.231695	0	23.42	<.0001
10	time2	10	Estimate	-0.155612	0.008792	-0.1731	-0.1382	99.413	-0.163448	-0.148571	0	-17.70	<.0001

This approach also yields valid marginal inference under MAR, provided the imputation model is correctly specified.

A direct comparison of IPW-GEE and MI-GEE was performed by examining both pooled parameter estimates and predicted marginal probabilities over time. As shown in Figure 1, the predicted marginal probabilities over time obtained from IPW-GEE and MI-GEE are virtually identical, confirming that the marginal conclusions are robust to the choice of MAR-based missing-data handling strategy. Apparent sign differences in individual polynomial coefficients reflect differences in parameterization rather than substantive differences in the fitted mean function.

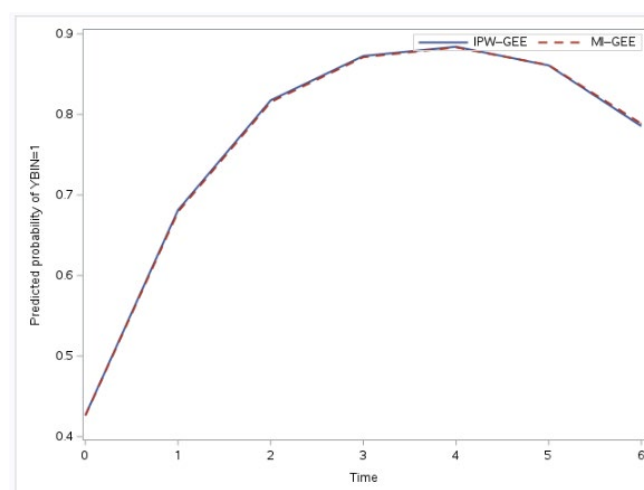


Figure 1. Predicted marginal probability over time obtained IPW-GEE (solid line) and MI-GEE (dashed line) showing near-identical trajectories.

Conclusion

In summary, MAR-valid inference was obtained for all outcomes using modeling strategies appropriate to each framework. The continuous outcome was analyzed using a likelihood-based LMM, and the binary outcome using both a likelihood-based GLMM and MAR-corrected marginal models. The agreement between IPW-GEE and MI-GEE, as well as consistency with likelihood-based analyses, provides strong evidence that the substantive conclusions are not driven by the handling of missing data under MAR.